

Section: Particles and Waves

Topic: Forces on Charged Particles

A beam of protons enters a region where there is a **uniform electric field** between two charged parallel plates. The beam is deflected as shown.

(a) State the direction of the electric field between the plates.

✔ Since protons are **positively charged**, they are deflected **toward the negative plate**. So the electric field must be directed **downward**, from the **positive** plate (top) to the **negative** plate (bottom).

(b) The electric field strength is $3.0 \times 10^3 \text{ V m}^{-1}$.

Calculate the force on a proton in the field.

Use:

$$F = Eq$$

Where:

- $E = 3.0 \times 10^3 \text{ V m}^{-1}$
- $q = 1.6 \times 10^{-19} \text{ C}$ (charge of a proton)

$$F = (3.0 \times 10^3)(1.6 \times 10^{-19}) = 4.8 \times 10^{-16} \text{ N}$$

(c) State what happens to the vertical acceleration of a proton as it moves through the field.

✔ The electric field is **uniform**, and the proton's **charge and mass are constant**, so:

$$a = \frac{F}{m}$$

Since F is constant, the **acceleration remains constant** throughout.

🧠 Revision Tips

- Field direction: from **positive to negative**
- For charged particles in an electric field:

$$F = Eq$$

- Acceleration of a charged particle is constant if the field is **uniform**
- Electric fields exert **forces on charges**, not masses